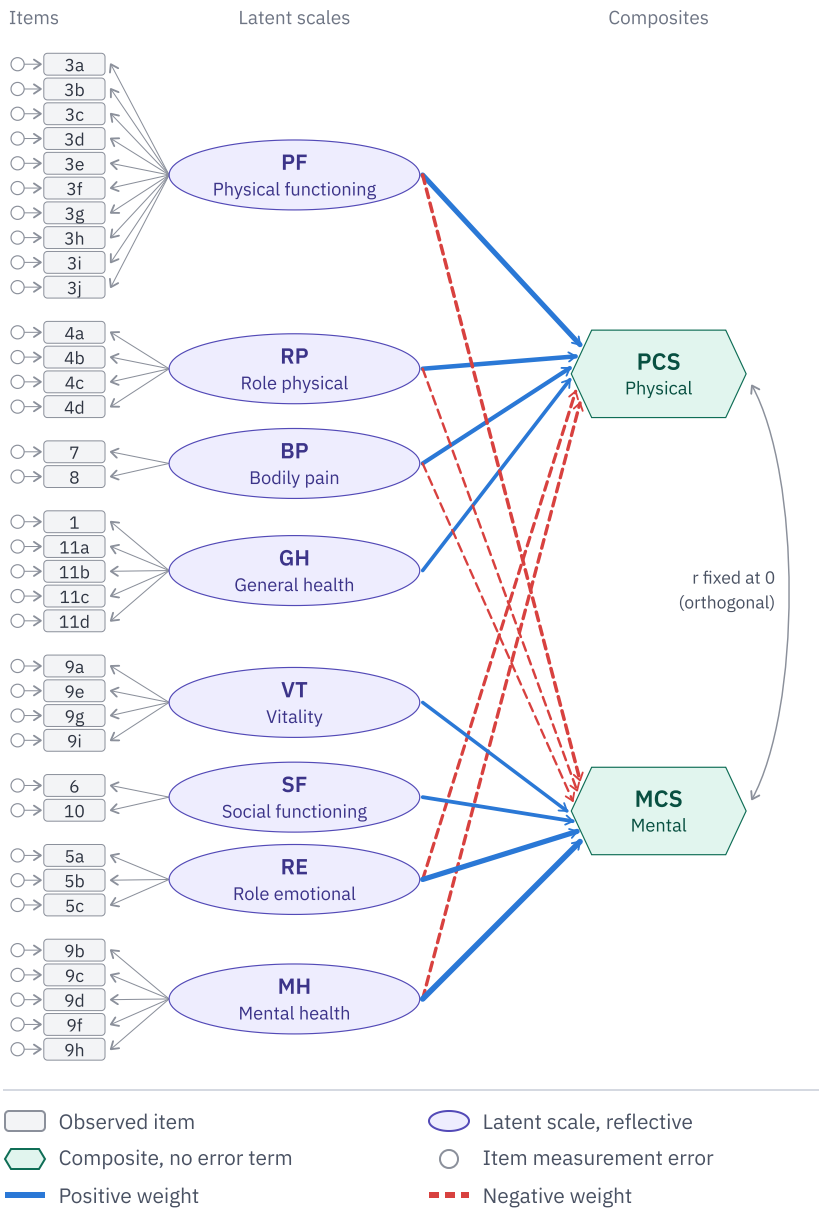


The SF-36 as a measurement model

Thirty-five questions measure eight latent dimensions of health. Those eight scales are then combined, with signed weights, into two summary indices of physical and mental health.



Line width is proportional to the size of the weight; weights under 0.05 in absolute value are not drawn. Covariances among the eight scales are estimated in a CFA but omitted here.

Reading the arrows. Scale-to-item arrows say the construct causes the answers, so the items should correlate with each other. Scale-to-summary arrows say the index is defined by the scales, so no fit test applies.

Why the negative weights. Physical and mental health are correlated, but the two summaries are forced to be uncorrelated. That makes PCS physical health net of mental health, and vice versa.

Physical scales

Physical functioning PF

Limits on physical activity, from bathing and dressing up to vigorous exercise. Ten items; the heaviest weight on PCS.

Role physical RP

Problems with work or daily activities caused by physical health: less time, less done, harder to do. Four items.

Bodily pain BP

How much pain the person has and how much it interferes with normal work. Two items.

General health GH

The person's overall rating of their health, now and looking ahead. Five items; almost all of its weight goes to PCS.

Mental scales

Vitality VT

Energy versus fatigue. Four items. Conceptually it straddles physical and mental health, but the orthogonal solution assigns it to MCS.

Social functioning SF

How much, and how often, physical or emotional problems get in the way of social life. Two items.

Role emotional RE

Problems with work or daily activities caused by emotional problems. Three items.

Mental health MH

Distress and well-being: nervousness, low mood, calm, and happiness. Five items; the heaviest weight on MCS.

Summary indices

Physical component summary PCS

A weighted sum of all eight standardized scales, scaled to mean 50 and SD 10 in the US population. Weights are positive on the physical scales and negative on RE and MH.

Mental component summary MCS

The mirror image: positive weights on the mental scales, negative weights on PF, RP, and BP.

Items and scoring

Item content is abbreviated. Items marked **R** are reverse-coded so that a higher score always means better health.

SF-36 version 1 numbering

Questions by scale

Physical functioning

3 options

- 3a Vigorous activities
- 3b Moderate activities
- 3c Lifting or carrying groceries
- 3d Climbing several flights of stairs
- 3e Climbing one flight of stairs
- 3f Bending, kneeling, or stooping
- 3g Walking more than a mile
- 3h Walking half a mile
- 3i Walking one hundred yards
- 3j Bathing or dressing yourself

Role physical

Yes/no (v2: 5 options)

As a result of physical health:

- 4a Cut down time spent on work
- 4b Accomplished less than you would like
- 4c Limited in the kind of work
- 4d Difficulty performing the work

Bodily pain

6 and 5 options

- 7 Pain magnitude **R**
- 8 Pain interference with work **R**

General health

5 options

- 1 Overall rating of general health **R**
- 11a I seem to get sick more easily than others
- 11b I am as healthy as anyone I know **R**
- 11c I expect my health to get worse
- 11d My health is excellent **R**

Vitality

6 options (v2: 5)

- 9a Felt full of life **R**
- 9e Had a lot of energy **R**
- 9g Felt worn out
- 9i Felt tired

Social functioning

5 options

- 6 Extent social activities were limited **R**
- 10 How often social activities were limited

Role emotional

Yes/no (v2: 5 options)

As a result of emotional problems:

- 5a Cut down time spent on work
- 5b Accomplished less than you would like
- 5c Did work less carefully than usual

Mental health

6 options (v2: 5)

- 9b Been a very nervous person
- 9c Felt so down that nothing could cheer you up
- 9d Felt calm and peaceful **R**
- 9f Felt downhearted and low
- 9h Been a happy person **R**

Item 2, the one-year health transition question, is not part of any scale.

Constructing the index

1 Recode the items

Reverse the R items so that higher values mean better health for every item.

2 Score each scale

Average the answered items in scale k . Score the scale only if at least half its items were answered.

$$R_k = (1 / n_k) \sum_{j \in k} x_j$$

3 Rescale to 0–100

Stretch each scale to its possible range, so 0 is the worst and 100 the best possible score.

$$X_k = 100 \times (R_k - \min_k) / (\max_k - \min_k)$$

4 Standardize against population norms

Use the US general-population means and standard deviations in the table below.

$$z_k = (X_k - \mu_k) / \sigma_k$$

5 Take the weighted sums

Every scale feeds both summaries. Weights are factor score coefficients from an orthogonal principal components solution.

$$PCS^* = \sum_k w_k^P z_k \quad MCS^* = \sum_k w_k^M z_k$$

6 Convert to T-scores

Rescale to mean 50 and SD 10. The same transformation gives norm-based scores for each scale.

$$PCS = 50 + 10 \cdot PCS^* \quad MCS = 50 + 10 \cdot MCS^* \quad T_k = 50 + 10 z_k$$

Norms and weights

Scale	μ	σ	w^P (PCS)	w^M (MCS)
Physical functioning	84.52	22.89	0.424	-0.230
Role physical	81.20	33.80	0.351	-0.123
Bodily pain	75.49	23.56	0.318	-0.097
General health	72.21	20.17	0.250	-0.016
Vitality	61.05	20.87	0.029	0.235
Social functioning	83.60	22.38	-0.008	0.269
Role emotional	81.29	33.03	-0.192	0.434
Mental health	74.84	18.01	-0.221	0.486

Worked example: average physical scales with RE and MH at $z = -2$ gives $PCS^* = (-0.192)(-2) + (-0.221)(-2) = 0.83$, so PCS is about 58. Poor mental health raises the physical score.

SF-36 version 1 constants from Ware, Kosinski & Keller (1994), *SF-36 Physical and Mental Health Summary Scales: A User's Manual* (1990 US norms). Verify them against the manual before scoring real data; SF-36v2 uses different norms and weights. The RAND 36-Item Health Survey 1.0 uses the same items and is free to use. Oblique (correlated) summary weights: Farivar, Cunningham & Hays (2007).